

Subject Description Form

Subject Code	EIE4108
Subject Title	Distributed Systems and Cloud Computing
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	EIE3320 Object Oriented Design and Programming
Objectives	This subject will provide students with the principles of distributed systems and cloud computing. It enables students to master the development skills to deliver and construct distributed services on the Web and cloud. Through a series of lab exercises, students will be able to develop interoperable and distributed Web and cloud applications.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the concepts of distributed systems and cloud computing. 2. Identify the key components in distributed systems and cloud services. 3. Build distributed systems. 4. Understand the advantages and limitations of different distributed systems and cloud architectures. 5. Understand the enabling technologies for building distributed systems. 6. Understand the different components of distributed systems. 7. Set up and configure a distributed application. <u>Category B: Attributes for all-roundedness</u> 8. Think critically. 9. Learn independently. 10. Work in a team and collaborate effectively with others. 11. Present ideas and findings effectively.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Introduction to Distributed Systems and Cloud Computing</u> 1.1. Definition and Examples of Distributed Systems; 1.2. Technologies for Network-Based Systems: multi-core and multi-threading; 1.3. Distributed and Cloud Computing Models: client-server; clusters; grids; peer-to-peer; remote procedure call; remote method invocation 1.4. Enabling Technologies: Socket programming; datagram sockets; stream-mode sockets 2. <u>Service-Oriented Architecture for Distributed Computing</u> 2.1. Service and Service-Oriented Architectures 2.2. RESTful Web Services: architectural principles of REST; AJAX; RESTful implementation 3. <u>Cloud Platform Architecture and Programming Environments</u> 3.1. Cloud Concepts Overview

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:	
	Specific Assessment Methods/Tasks	Remark
	Short quizzes	Short multiple-choice quizzes are conducted to measure the students' understanding of the theories and concepts as well as their comprehension of subject materials.
	Assignments and tests	Assignments are programming exercises demonstrating the operating principles of different distributed systems and cloud computing. The purposes are to strengthen students' understanding of the topics they learnt in classes. Students will be assessed based on their ability to apply concepts and skills learnt in the classroom. Students need to think critically and creatively in order to come up with an alternate solution for an existing problem. Tests are given to students to assess their competence level of knowledge and comprehension and their ability to apply knowledge and skills in new situations.
	Laboratory sessions and lab reports	Students are required to build distributed systems and cloud computing applications during the lab sessions. They are also required to write reports to explain the architecture and operating principles of their systems. Students will be assessed based on (1) their ability to apply knowledge that they learn in classes to build distributed systems and cloud computing applications, and (2) their ability to write a clear report that explains the principle of operation and architecture of the systems that they have created.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture/Tutorial/Practice Classes	30 Hours
	• Laboratory Classes	9 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours

Reading List and References	References: 1. John Culkin and Mike Zazon. <i>AWS Cookbook: Recipes for Success on AWS</i>. O'Reilly Media, 2022. 2. I. Foster and D.B. Gannon, <i>Cloud Computing for Science and Engineering</i>”, MIT Press, 2017. 3. T. Erl et al. <i>SOA with REST: Principles, Patterns & Constraints for Building Enterprise Solutions with REST</i>, Prentice Hall 2013. 4. G. Coulouris, <i>Distributed Systems: Concepts and Design</i>, 5th ed., Addison-Wesley, 2011. 5. T. Erl, <i>Cloud Computing: Concepts, Technology and Architecture</i>, Prentice-Hall, 2013. 6. A. Mishra, <i>Machine Learning in the AWS Cloud: Add Intelligence to Applications with Amazon SageMaker and Amazon Rekognition</i>, 1st edition. Newark: John Wiley & Sons, Incorporated, 2019.
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